

Alpha-Szegő Rank-One Dynamics: Exact Reduction and Global Compactness

HCS-C386 theorem and reproducibility package

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Abstract

We reconstruct the complete rank-one rational dynamics of the perturbed cubic Szegő equation with its original physical time. An exact Fourier convolution gives a six-real-variable system preserving mass, momentum and energy. A logarithmic radial bound excludes finite-time loss of the rank-one condition. The conserved energy defect supplies an explicit lower bound on the distance of the pole from the unit circle whenever the defect is nonzero. Every negative-perturbation orbit is therefore relatively compact. At zero perturbation the exceptional zero-defect data are degree-one inner functions with a pure phase evolution. Spatially constant data form a separate rank-zero face, where the energy equality alone cannot diagnose growth. This first round closes the nonlinear definition, global existence, and the compactness and degeneracy controls needed for a sharp cascade theorem. The model and its classical threshold lineage are credited; this is a source-theorem reconstruction, not a claim of a newly discovered equation.

中文摘要

本文在原始物理时间下重建带线性扰动的三次塞戈方程的完整秩一有理动力学。

精确傅里叶卷积给出六个实变量的闭合方程，并保持质量、动量与能量。

径向变量的对数导数估计排除有限时间离开秩一流形。

非零能量缺陷给出极点距单位圆的定量下界，因此相应轨道相对紧。

负扰动的全部轨道均紧；零扰动的零缺陷数据恰为一次内函数，只有纯相位演化。

空间常数是单列的秩零边界，不能用能量等式判断其增长。本轮完成全局定义与退化控制。

Keywords: alpha-Szegő; rank-one dynamics; energy threshold; turning point; Sobolev cascade; isospectrality

关键词：塞戈方程；秩一动力学；能量阈值；转折点；索伯列夫级联；等谱性

1 One invariant nonlinear source

A finite-rank invariant manifold need not be a finite Fourier approximation. Here it contains infinitely many modes and an exact nonlinear evolution. The cubic Szegő equation and its Hankel structure originate in Gérard–Grellier [1]; Xu [2] established the rank-one threshold for its linear perturbation. Our contribution within this repository is a complete convention-preserving proof and reproducibility package. In particular, we retain the rank hypothesis, the turning point, and the distinction between an auxiliary determinant and physical evolution. We do not claim literature priority for the classical threshold.

Use normalized circle measure and let Π retain nonnegative Fourier modes. For every real α fix

$$iu_t = \Pi(|u|^2 u) + \alpha(u \mid 1), \quad u(z) = b + \frac{cz}{1-pz}, \quad c \neq 0, \quad |p| < 1. \quad (1)$$

The phase space in (1) is $\mathcal{L}(1)$; constants will be treated separately. The physical time is t , without rescaling. Set

$$d = 1 - |p|^2, \quad \rho = |c|, \quad B = |b|^2, \quad Q = B + \frac{\rho^2}{d}, \quad M = \frac{\rho^2}{d^2} > 0. \quad (2)$$

Energy and its defect are

$$E_\alpha = \frac{1}{4} \|u\|_4^4 + \frac{\alpha}{2} |b|^2, \quad \delta = E_\alpha - \frac{Q^2}{4} - \frac{\alpha Q}{2}. \quad (3)$$

We always take the positive square root of M . On a trajectory, $\rho = \sqrt{M}d$ and $B = Q - Md$.

Theorem 1 (Exact all-mode reduction and global existence). *Every initial datum in $\mathcal{L}(1)$ has a unique solution for all real time. Its coefficients satisfy*

$$\begin{aligned} i\dot{b} &= (B + 2\rho^2/d + \alpha)b + \rho^2 c\bar{p}/d^2, \\ i\dot{c} &= (2B + \rho^2/d^2)c + 2b\rho^2 p/d, \\ i\dot{p} &= c\bar{b} + \rho^2 p/d. \end{aligned} \quad (4)$$

The quantities Q, M, E_α are conserved. These statements hold at $b = 0$ and $p = 0$ and involve no Fourier cutoff.

Proof. The geometric Fourier coefficients are $u_0 = b$ and $u_n = cp^{n-1}$ for $n \geq 1$. Their absolute convergence on strict subdisks justifies convolution in $\Pi(|u|^2 u)$. Compare the coefficients of $1, z/(1-pz), z^2/(1-pz)^2$; the derivative of the last coordinate is $c\dot{p}$, not merely $\dot{p}$. The result is (4). A direct quartic sum gives

$$4E_\alpha = B^2 + \frac{4B\rho^2}{d} + \frac{\rho^4(1+|p|^2)}{d^3} + \frac{4\rho^2\Re(bp\bar{c})}{d^2} + 2\alpha B. \quad (5)$$

Differentiating (2) and (5) along (4) cancels every term, giving the three conservation laws.

To justify all-time continuation, put $X = bp\bar{c}$. Direct differentiation gives

$$\dot{d} = 2\Im X, \quad \dot{\rho} = 2\sqrt{M}\Im X, \quad |\dot{\rho}/\rho| \leq 2\sqrt{QM}. \quad (6)$$

Hence $\rho(t) \geq \rho(0)e^{-2\sqrt{QM}|t|}$ on every finite time interval. The conserved quantities bound $|b| \leq \sqrt{Q}, \rho \leq \sqrt{M}$, and $d = \rho/\sqrt{M}$ away from zero on that interval. The real six-dimensional vector field is locally Lipschitz on $d > 0$, so its continuation theorem excludes finite-time exit. The resulting geometric series solves (1) in every H^s . This argument proves the named invariant-manifold theorem, not global well-posedness for arbitrary Hardy data. $\square$

2 Compactness and the exact exceptional faces

The defect is useful because it vanishes precisely where a compactness estimate could fail.

Lemma 1 (Factored defect and a quantitative bound). *For every trajectory in $\mathcal{L}(1)$,*

$$\delta = \frac{Md}{2} \left(\left| b + \frac{c\bar{p}}{d} \right|^2 - \alpha \right). \quad (7)$$

If $\delta \neq 0$, then for every real t ,

$$d(t) \geq \frac{2|\delta|}{M((\sqrt{Q} + \sqrt{M})^2 + |\alpha|)} > 0. \quad (8)$$

The orbit is relatively compact in $\mathcal{L}(1)$ and bounded in every H^s .

Proof. Subtract the two terms in (3) from (5); the remaining quadratic expression is (7). Since $|b + c\bar{p}/d| \leq \sqrt{Q} + \sqrt{M}$, taking absolute values proves (8). Thus $|p|$ lies in a strict subdisk and $|c| = \sqrt{M}d$ stays positive, while b, c remain bounded. These inequalities give compact closure within the coefficient manifold. Every Sobolev norm is then bounded by a convergent geometric series. $\square$

For $\alpha < 0$, (7) has strictly positive defect; thus all rank-one orbits are relatively compact. On a compact set of initial data, the positive continuous defect has a positive minimum, making the bound uniform for that set. This does not assert uniformity across data approaching a positive-alpha zero-defect hypersurface.

For $\alpha = 0$, nonzero defect is already settled. Zero defect forces $b = -c\bar{p}/d$. Writing $A = c/d$ yields

$$Q = M = |A|^2, \quad u(z) = A \frac{z - \bar{p}}{1 - pz}, \quad u(t, z) = e^{-iQt} u(0, z). \quad (9)$$

The quotient in (9) has modulus one on the circle. Substitution in (4) gives $\dot{p} = 0, \dot{b} = -iQb$ and $\dot{c} = -iQc$, proving the formula even at $p = 0$. Therefore every zero-alpha rank-one orbit is also relatively compact.

Rank zero cannot inherit the threshold. If $c = 0$, the parameter p is redundant and the complete solution is

$$u(t, z) = b(0)e^{-i(|b(0)|^2 + \alpha)t}. \quad (10)$$

Every constant satisfies $\delta = 0$ but has constant Sobolev norm. The zero function is stationary. A nonzero constant is stationary exactly when $|b|^2 + \alpha = 0$; otherwise its least temporal period is $2\pi/||b|^2 + \alpha|$. These facts explain why the rank-one hypothesis cannot be dropped from a positive-alpha growth criterion.

3 Evidence boundary and conclusion

Source face	Defect	Established behavior
Rank one, $\alpha < 0$	positive	relatively compact
Rank one, $\alpha = 0$	nonzero	relatively compact
Rank one, $\alpha = 0$	zero	inner-function phase orbit
Rank zero	always zero	constant-norm phase orbit

The regime table records a rank distinction, not a fitted classifier. All complex phases are retained in the coefficient ODE. The all-time and all-frequency conclusions are proved analytically; finite computations test independent reconstructions of their formulas.

The claim remains the rank-one source theorem. No arbitrary Hardy-data or higher-rank threshold, complete generic primitive census, target arithmetic local data, target Euler factors, root number, automorphy, target divisor, target functional equation, target-zero matching, or Hilbert–Pólya operator is claimed. Route B remains disabled. The firewall is NO_BAD_EULER_OR_ROOT_NUMBER.

Literature ownership and reproducibility. The classical equation and threshold belong to the cited sources. Our package makes the turning-point argument, rank-zero failure, physical clock and auxiliary-determinant limitation auditable together. No publication acceptance, worldwide novelty, external-human review or cross-provider validation is asserted. AI-assisted local drafting and independent collaborating-agent review are disclosed. The source proof, locked evaluation, independent scripts and three substantive PDF revisions accompany the article.

Conclusion. Round zero: exact nonlinear reduction and compactness. The rational manifold closes the full coefficient dynamics and every nonpositive-alpha and constant boundary, preparing a valid threshold question.

References

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